Chapter 12: Electricity

1. Introduction

Electricity has an important place in modern society. It is a controllable and convenient form of energy used in homes, schools, hospitals, and industries. This chapter introduces the core concepts of electricity, its circuits, and how it is measured.

2. Electric Current and Circuit

If electric charge flows through a conductor (for example, through a metallic wire), we say that there is an **electric current** in the conductor. A continuous and closed path of an electric current is called an **electric circuit**.

- **Electric Current (I):** The rate of flow of electric charges.
  Formula: $I = Q / t$ (Where I is current, Q is charge, and t is time).
- **Unit of Current:** Ampere (A). Specifically, 1 Ampere = 1 Coulomb / 1 Second.
- **Measurement:** Ammeter. It is always connected in series in a circuit.

3. Electric Potential and Potential Difference

For electrons to move in a conductor, there must be a difference in electric pressure, called the **potential difference**. This is typically provided by a cell or a battery.

- **Potential Difference (V):** Work done to move a unit charge from one point to another.
  Formula: $V = W / Q$ (Where V is potential difference, W is work done, and Q is charge).
- **Unit:** Volt (V).
- **Measurement:** Voltmeter. It is always connected in parallel across the points where the potential difference is to be measured.

4. Ohm's Law

Proposed by Georg Simon Ohm, this law gives a relationship between the current flowing through a conductor and the potential difference across its ends.

> **Ohm's Law:** At constant temperature, the current flowing through a conductor is directly proportional to the potential difference across its ends.
>
> **Formula:** $V \propto I \Rightarrow V = I * R$

Here, **R** is the **Resistance** of the conductor. Resistance is the property of a conductor to resist the flow of charges through it.
Unit of Resistance: Ohm (Ω).

Factors on which Resistance depends

1. **Length (l):** Resistance is directly proportional to the length ($R \propto l$).
2. **Area of cross-section (A):** Resistance is inversely proportional to the area of cross-section ($R \propto 1/A$). Thicker wires have less resistance.
3. **Nature of the material.**
4. **Resistivity (ρ):** $R = \rho(l/A)$. Resistivity is a characteristic property of the material dependent on temperature.

5. Resistors in Series and Parallel

Series Combination

When two or more resistors are connected end to end consecutively.
- The current is the same in every part of the circuit.
- The total potential difference is the sum of individual potential differences ($V = V1 + V2 + V3$).
- **Equivalent Resistance (Rs):** $Rs = R1 + R2 + R3$.

Parallel Combination

When two or more resistors are connected across two common points.
- The potential difference is the same across each resistor.
- The total current is the sum of separate currents through each resistor ($I = I1 + I2 + I3$).
- **Equivalent Resistance (Rp):** $1/Rp = 1/R1 + 1/R2 + 1/R3$.

6. Heating Effect of Electric Current

If an electric circuit is purely resistive, the source energy continually gets dissipated entirely in the form of heat. This is known as the heating effect of electric current.

> **Joule's Law of Heating:** The heat produced in a resistor is directly proportional to the square of current (I^2), resistance (R), and time (t).
>
> **Formula:** $H = I^2Rt$

Applications: Electric iron, electric toaster, electric oven, electric heater, and electric fuse.

7. Electric Power

The rate at which electric energy is dissipated or consumed in an electric circuit.

- **Formula:** $P = V * I$ (also $P = I^2R$ or $P = V^2/R$).
- **Unit:** Watt (W).
- **Commercial Unit of Energy:** Kilowatt-hour (kWh), commonly known as 'unit'.
  1 kWh = 1000 W * 3600 s = 3.6×10^6 Joules.

8. Key Points / Important Notes

- An Ammeter has very low resistance so it does not affect the current of the circuit. A Voltmeter has very high resistance so it doesn't draw much current.
- **Alloys** like Nichrome or Constantan are generally used in electrical heating devices because their resistivity is higher than pure metals and they do not oxidize (burn) readily at high temperatures.
- Electric bulbs are usually filled with chemically inactive nitrogen and argon gases to prolong the life of the filament (usually made of Tungsten, which has a very high melting point).

9. Frequently Asked Questions

Q. Why is the series arrangement not used for domestic circuits?

Ans. In a series circuit, if one appliance fails, the circuit is broken and all appliances stop working. Also, different appliances need different currents to operate properly, which a series circuit cannot provide because current is constant throughout.

Q. How does use of a fuse wire protect electrical appliances?

Ans. A fuse is a wire of low melting point inserted in the live wire. If a current larger than the specified value flows, the fuse wire heats up, melts, and breaks the circuit, thus preventing damage to appliances.

10. Multiple Choice Questions (MCQs)

1. The unit of electric potential difference is:

a. Joule

b. Ampere

c. Volt

d. Ohm

Answer: c) Volt

2. By convention, the direction of conventional current is taken as:

a. the direction of flow of electrons

b. opposite to the direction of flow of electrons

c. from negative to positive terminal

d. None of these

Answer: b) opposite to the direction of flow of electrons

3. Ammeter is always connected in _____ in a circuit.

a. Parallel

b. Series

c. Either series or parallel

d. Depends on the resistance

Answer: b) Series

4. The resistance of a conductor is directly proportional to its:

a. Length

b. Area of cross-section

c. Temperature

d. Resistivity

Answer: a) Length

5. According to Ohm's law, if we double the potential difference across a resistor, the current will:

a. Halve

b. Double

c. Become four times

d. Remain same

Answer: b) Double

6. If two resistors of 5 ohms and 10 ohms are connected in series, their equivalent resistance is:

a. 50 ohms

b. 15 ohms

c. 3.33 ohms

d. 2 ohms

Answer: b) 15 ohms

7. Two unequal resistors are connected in parallel. Which of the following is true?

a. Current is same in both

b. Potential difference is same across both

c. Both current and potential difference are same

d. Equivalent resistance is higher than the individual resistances

Answer: b) Potential difference is same across both

8. Which of the following represents electrical power?

a. I^2R

b. IR^2

c. VI^2

d. V^2I

Answer: a) I^2R

9. Commercial unit of electrical energy is:

a. Joule

b. Coulomb

c. Kilowatt

d. Kilowatt-hour (kWh)

Answer: d) Kilowatt-hour (kWh)

10. 1 kWh is equal to:

a. 3.6×10^3 Joules

b. 3.6×10^6 Joules

c. 36×10^6 Joules

d. 3.6 Joules

Answer: b) 3.6×10^6 Joules

11. A fuse wire should have:

a. High resistance and high melting point

b. Low resistance and low melting point

c. High resistance and low melting point

d. Low resistance and high melting point

Answer: c) High resistance and low melting point

12. The filament of an electric bulb is made of:

a. Copper

b. Tungsten

c. Aluminium

d. Nichrome

Answer: b) Tungsten

13. The rate of flow of electric charges is called:

a. Electric potential

b. Electric current

c. Electric power

d. Electric field

Answer: b) Electric current

14. Which relation defines Joule's law of heating?

a. $H = I^2Rt$

b. $H = VIt$

c. $H = V^2t/R$

d. All of these

Answer: d) All of these

15. The resistivity of a certain material is 1.6 x 10^-8 Ω m. The material is most likely a:

a. Good conductor

b. Semiconductor

c. Insulator

d. Superconductor

Answer: a) Good conductor (Metals have low resistivity)

16. If radius of a copper wire is doubled, its resistance becomes:

a. Double

b. Half

c. Four times

d. One-fourth

Answer: d) One-fourth (Area increases by 4 times, $R \propto 1/A$)

17. Which is the correct symbol for a variable resistance?

a. A zigzag line

b. A zigzag line with an arrow striking through it

c. A straight line

d. Two parallel lines

Answer: b) A zigzag line with an arrow striking through it

18. The equivalent resistance of three equal resistors connected in parallel is exactly equal to _____ of one resistor.

a. Three times

b. One-third

c. Same as

d. One-ninth

Answer: b) One-third

19. If you stretch a wire to make it longer, its resistance will:

a. Decrease

b. Increase

c. Remain same

d. Become zero

Answer: b) Increase

20. Work done moving a charge of 2C across two points having a potential difference 12 V is:

a. 24 Joules

b. 6 Joules

c. 12 Joules

d. 10 Joules

Answer: a) 24 Joules ($W = V \times Q = 12 \times 2$)

11. Short Answer Questions

1. Define 1 volt.

Answer: 1 volt is the potential difference between two points in a current-carrying conductor when 1 joule of work is done to move a charge of 1 coulomb from one point to the other.

2. State Ohm's law.

Answer: Ohm's law states that the physical conditions (temperature etc.) remaining constant, the current flowing through a metallic conductor is directly proportional to the potential difference applied across its ends ($V = IR$).

3. Why are coils of electric toasters and electric irons made of an alloy rather than a pure metal?

Answer: Alloys have a higher resistivity than that of their constituent pure metals. Furthermore, alloys do not oxidize or burn rapidly at high temperatures, making them highly suitable for heating devices.

12. Long Answer Questions

1. Explain Joule's Law of Heating. How is this effect utilized in an electric fuse to protect circuits?

Answer:
Joule's Law of heating states that the heat produced in a resistor is:
(i) Directly proportional to the square of current for a given resistance (I^2)
(ii) Directly proportional to resistance for a given current (R)
(iii) Directly proportional to the time for which current flows through the resistor (t).
This yields the equation: $H = I^2Rt$.

In an electric fuse: A fuse is a wire of defined length and low melting point connected in series with the live wire of an appliance. If a surge of inappropriately high current flows into the circuit, Joule heating increases significantly (since $H \propto I^2$). This rapid heat production melts the fuse wire, breaking the electrical circuit and protecting the expensive appliances and household wiring from damage or catching fire.

2. Compare the advantages of connecting electrical appliances in parallel over connecting them in series.

Answer:

1. Independent Operation: In a parallel circuit, if one appliance stops working due to some defect, all other appliances keep working normally. In a series circuit, if one appliance fails, the whole circuit breaks.

2. Different Current needs: Different appliances (like an AC and a bulb) require different amounts of current to operate. A parallel circuit divides the total current among the appliances matching their requirement, whereas series sends exactly the same current through all components.

3. Voltage Supply: In a parallel connection, every appliance gets the same potential difference (usually 220V) as that of the power supply terminal, ensuring they work at their maximum power capacity.